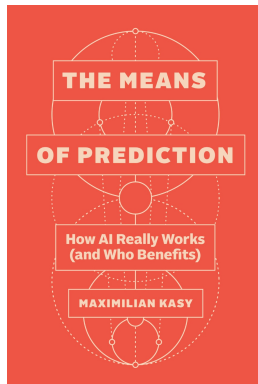


The Means of Prediction How AI Really Works (And Who Benefits)

Maximilian Kasy

Department of Economics, University of Oxford

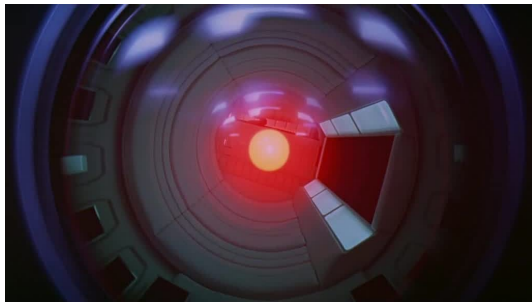
September 2025



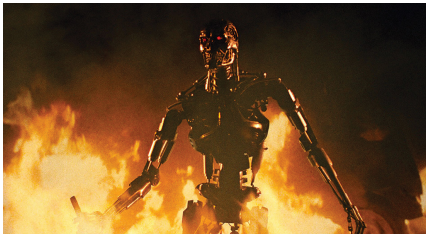
Are you scared of AI?

A popular dystopian story:

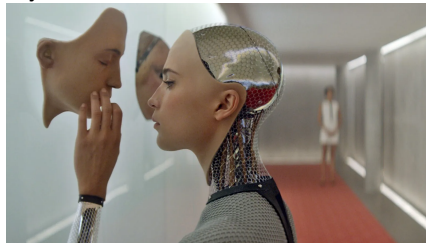
- AI will attain superhuman capabilities,
- will start to self-improve exponentially,
- and will threaten human existence in the name of self-preservation.



2001: A Space Odyssey



Terminator



Ex Machina

- Such stories touch on our deepest fears:
 - Losing our livelihoods, autonomy, lives, and loved ones,
 - to inscrutable and inevitable forces.
- But they don't enable good decisions:
 - Make it seem like AI and its use are fate.
 - Obscure conflicts over who controls AI.
- Intentional obfuscation by tech players?

A more accurate story

1. AI is automated decision-making using *optimization*.
2. Key issue: Who gets to pick the *objectives* that AI optimizes?
(Not: Did the AI fail to optimize?)
3. Power flows from control of AI *inputs*:
data, compute, expertise, energy.
4. We need *democratic control* of AI objectives
by those affected by AI decisions.

Some examples

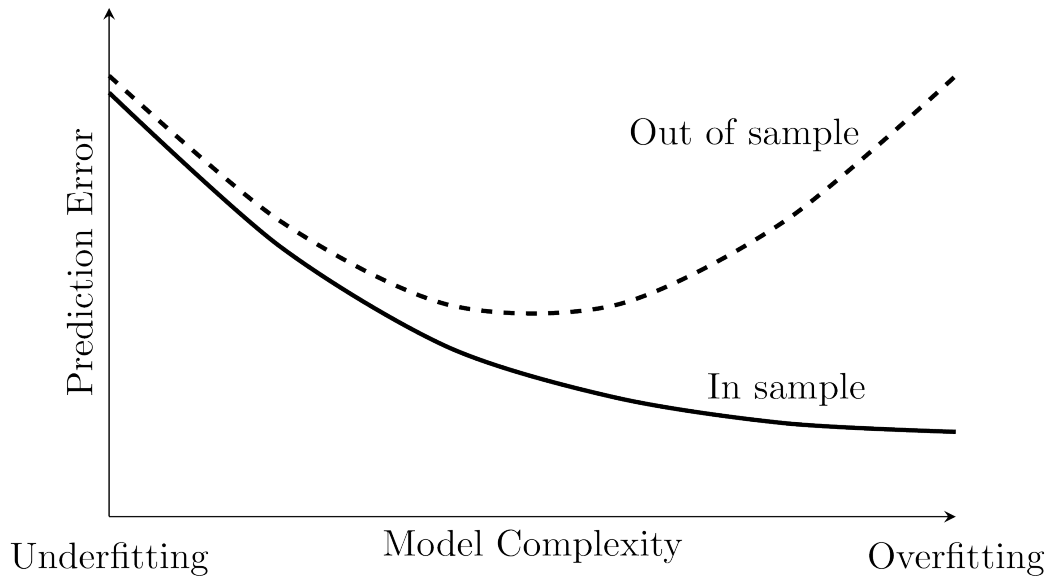
Beyond the headline-grabbing large language models:

- Algorithmic management of gig-workers.
- Automatic screening of job candidates.
- Filtering and selection of social media feeds, search engine results.
- Ad targeting.
- Predictive policing and incarceration.
- Automated choice of bombing/assassination targets (e.g. Gaza).

How AI works

The book explains the foundations of machine learning and AI *without math*:

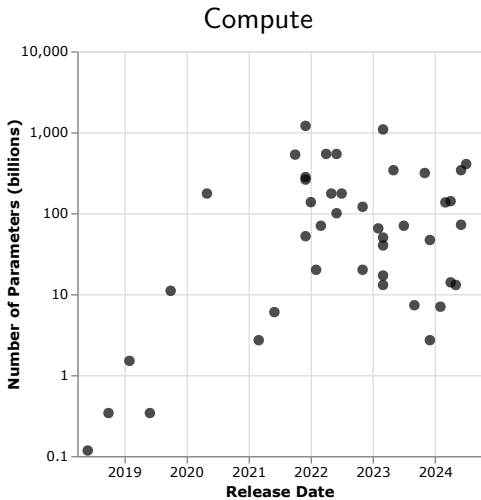
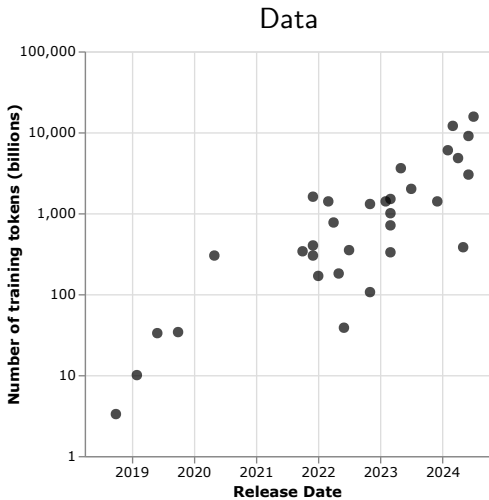
- AI is automated decision-making, maximizing some reward.
- Machine learning is AI using statistics.
- Supervised learning: Prediction
 - Overfitting versus underfitting, tuning.
 - Deep learning, transformers.
- Online learning: Choosing actions over time.
 - Exploration versus exploitation.
 - Planning.



The means of prediction

- These foundations clarify what resources are needed for AI:
 - Data
 - Compute
 - Expertise
 - Energy
- Implications:
 - Potential for future improvements (domain-dependent).
 - Control of AI by controlling its inputs.
 - Contests over property rights, externalities.

Scaling of large language models



Agents of change

Who can align AI objectives with social welfare?

- Interests, values, and strategic leverage.
- AI companies? Constrained by profit maximization.
- AI discourse should address others:
 - Workers (click-, gig-, tech-), consumers,
 - media and public opinion, state and law.
- Ultimate goal:
 - Democratic control of AI objectives
 - by those impacted by AI decisions.

Ideological obfuscation

- Ideology: Represents
 - Interests of a particular group as those of society at large.
 - Contingent choices as objective necessity.
 - Social relationships as technical ones.
- Popular AI stories that prevent change:
 1. *Man versus machine*: Obfuscates conflicts within society.
 2. *Intelligence explosion*: Not human choices but autonomous process.
 3. *Only experts understand AI*: Prevents democratic control of tech companies.
 4. *If we don't do it, China will*: Political inevitability.

Regulating algorithms

Ramifications of this perspective for various policy domains:

- Value alignment and the limits of AI
- Privacy and data ownership
- Workplace automation and the labor market
- Fairness and algorithmic discrimination
- Explainability of algorithms and algorithmic decisions

Thank you!

Book available for ordering here:

https:

`//press.uchicago.edu/ucp/books/
book/chicago/M/bo255887145.html`

